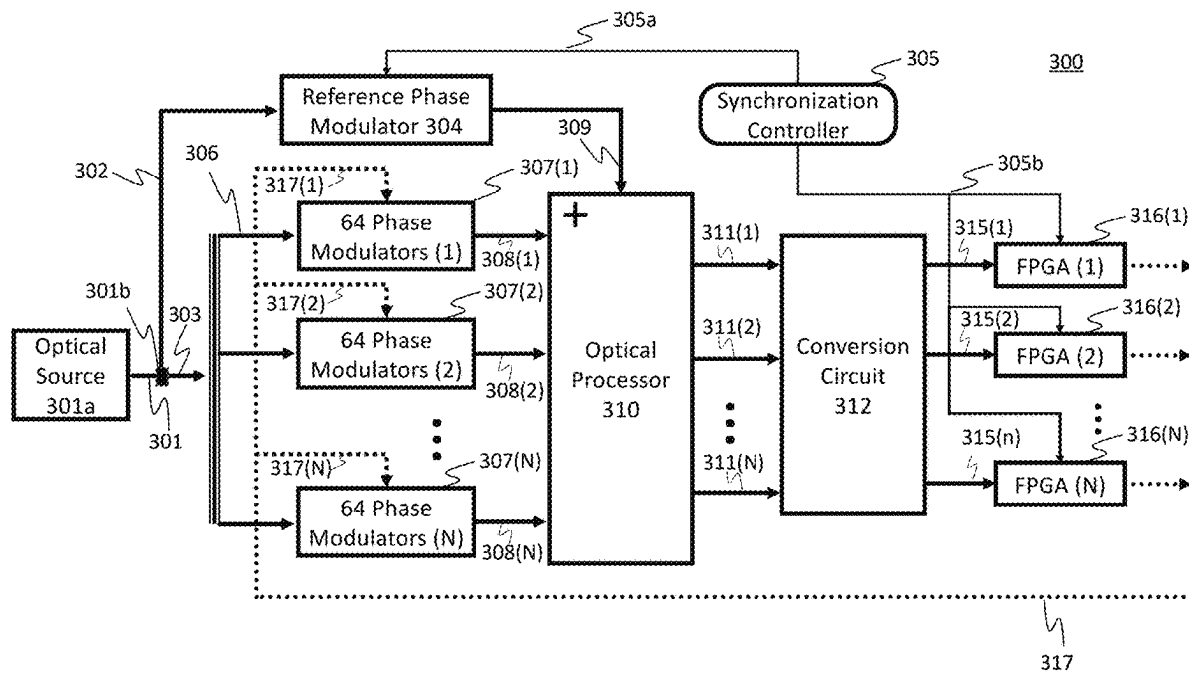




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(19) **United States**(12) **Patent Application Publication**
Hallak et al.(10) **Pub. No.: US 2025/0279579 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **WIDELY SCALABLE, MODULAR PHASE
CONTROL OF OPTICAL CHANNELS**(60) Provisional application No. 63/234,151, filed on Aug.
17, 2021.(71) Applicant: **Phase Sensitive Innovations, Inc.**,
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Matthew Gallion, Middletown, DE
(US)(52) **U.S. Cl.**
CPC **H01Q 3/2676** (2013.01); **H04B 10/6165**
(2013.01); **H04B 2210/006** (2013.01)(21) Appl. No.: **19/205,692**(22) Filed: **May 12, 2025****Related U.S. Application Data**(63) Continuation of application No. 18/616,753, filed on
Mar. 26, 2024, now Pat. No. 12,300,902, which is a
continuation of application No. 17/890,017, filed on
Aug. 17, 2022, now Pat. No. 11,942,695.(57) **ABSTRACT**

A plurality of digital processors may be used to adjust phases in a plurality of phase modulators. The plurality of digital processors may receive a periodic pulse, or heartbeat signal, from a synchronization controller in order to control the digital processors. The synchronization controller may output an additional signal used to determine and to control the phase of the signals output from the plurality of phase modulators.



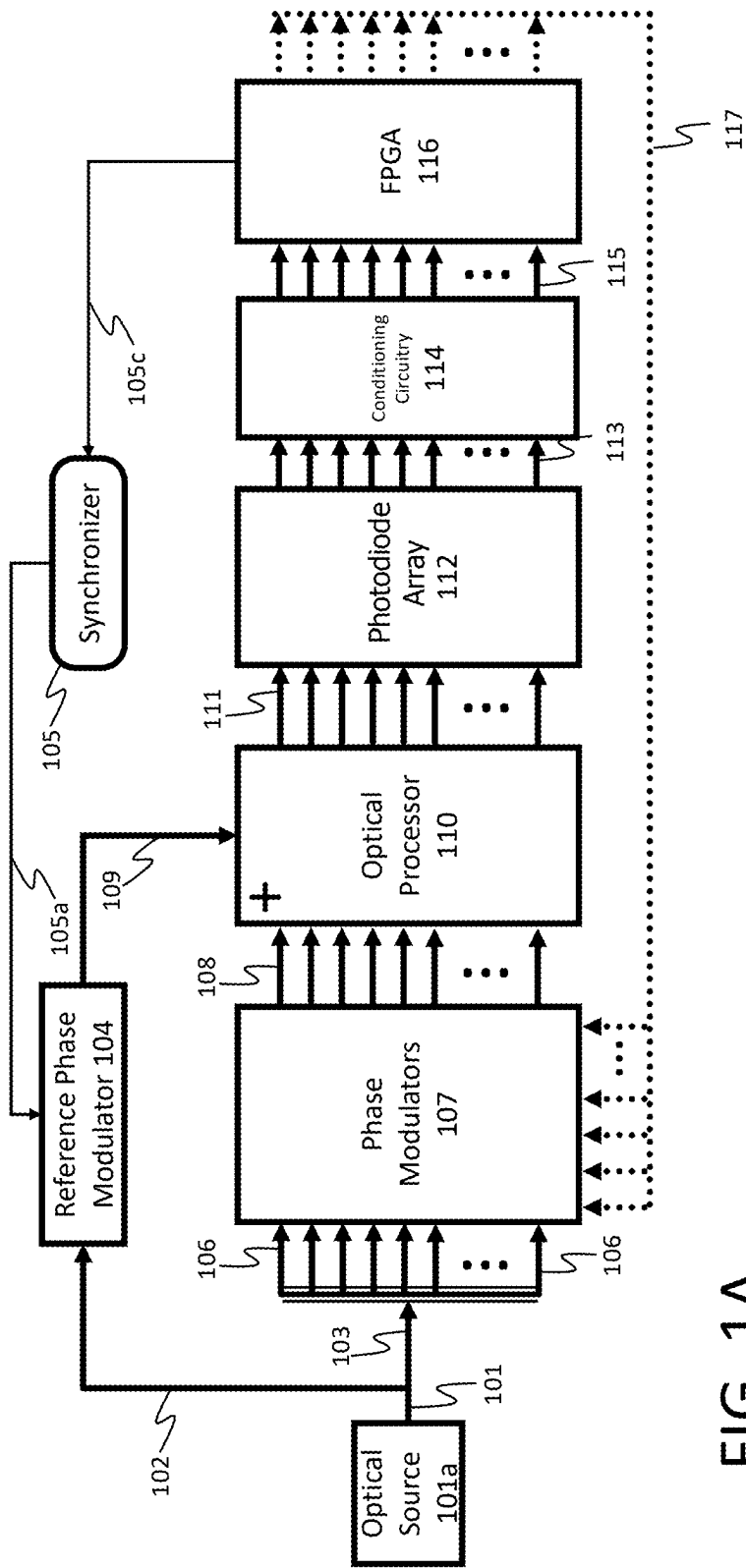


FIG. 1A

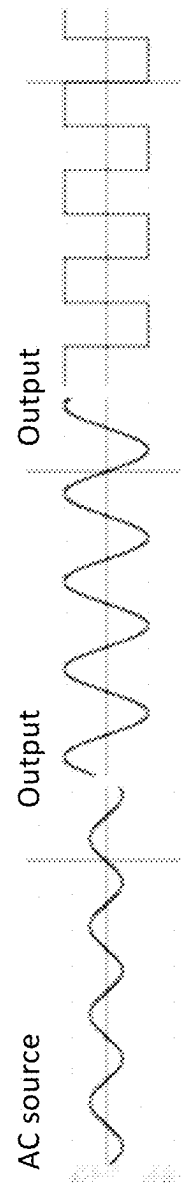


FIG. 1B

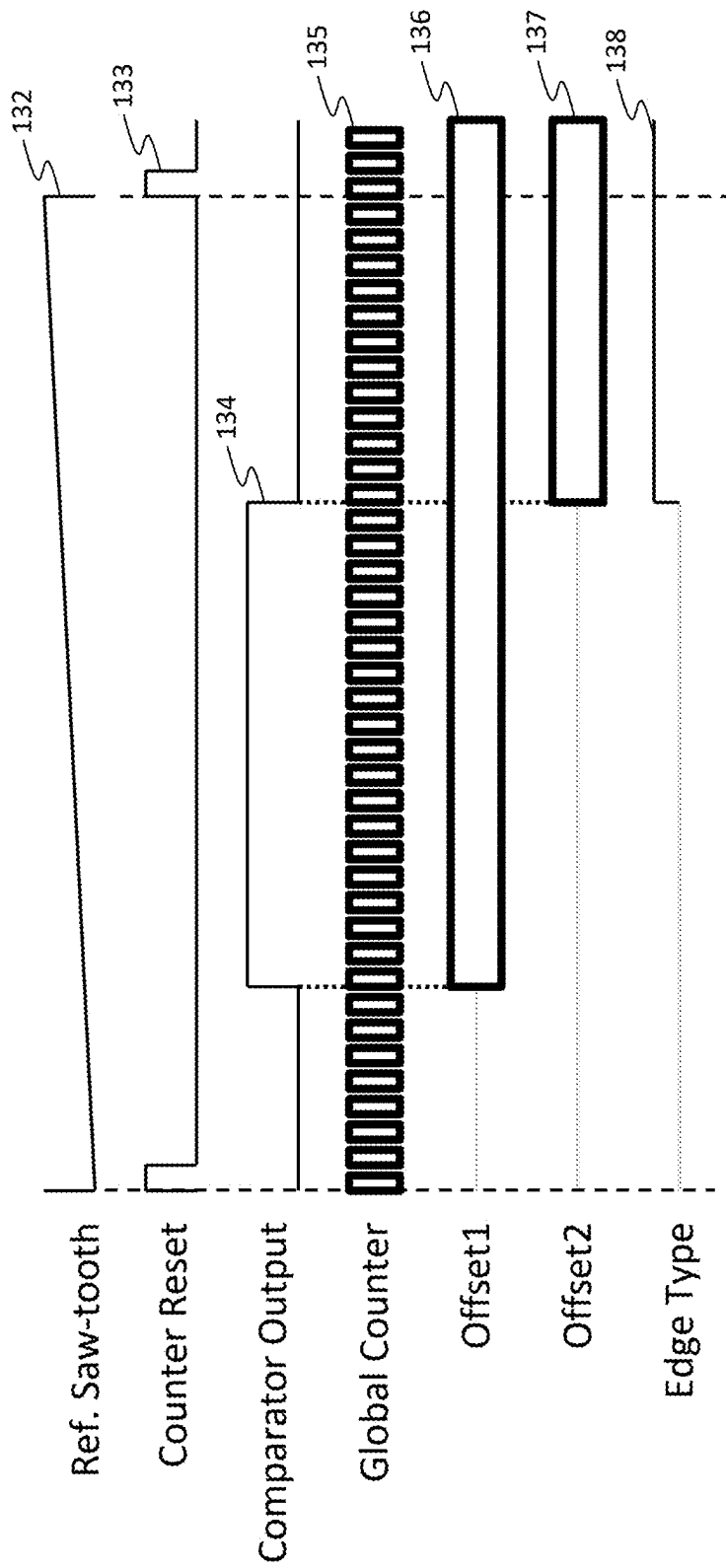


FIG. 1C

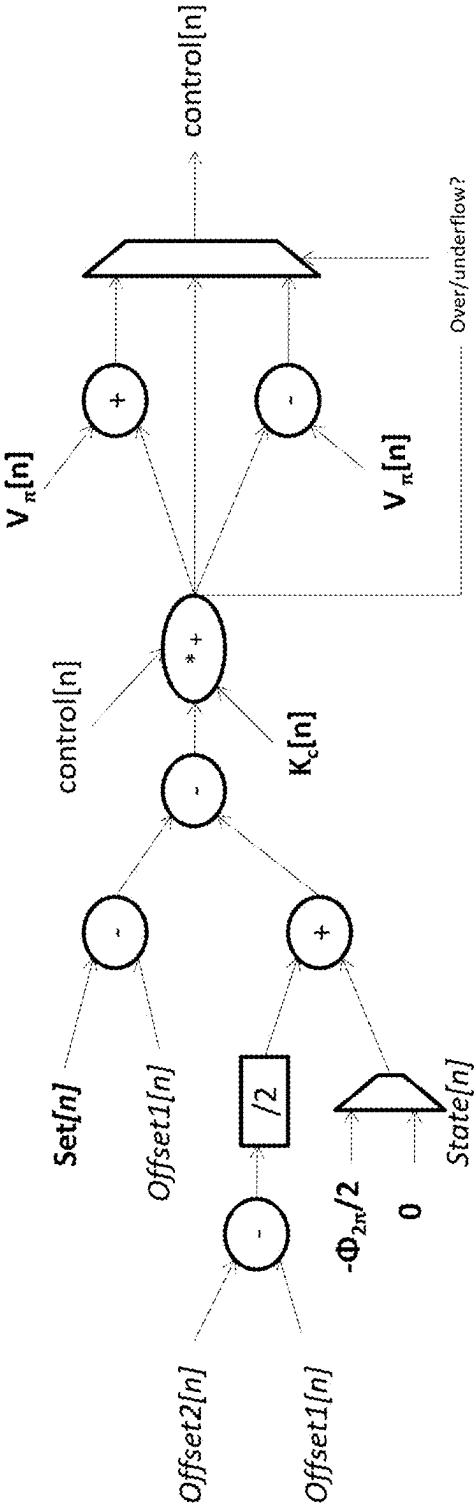


FIG. 1D

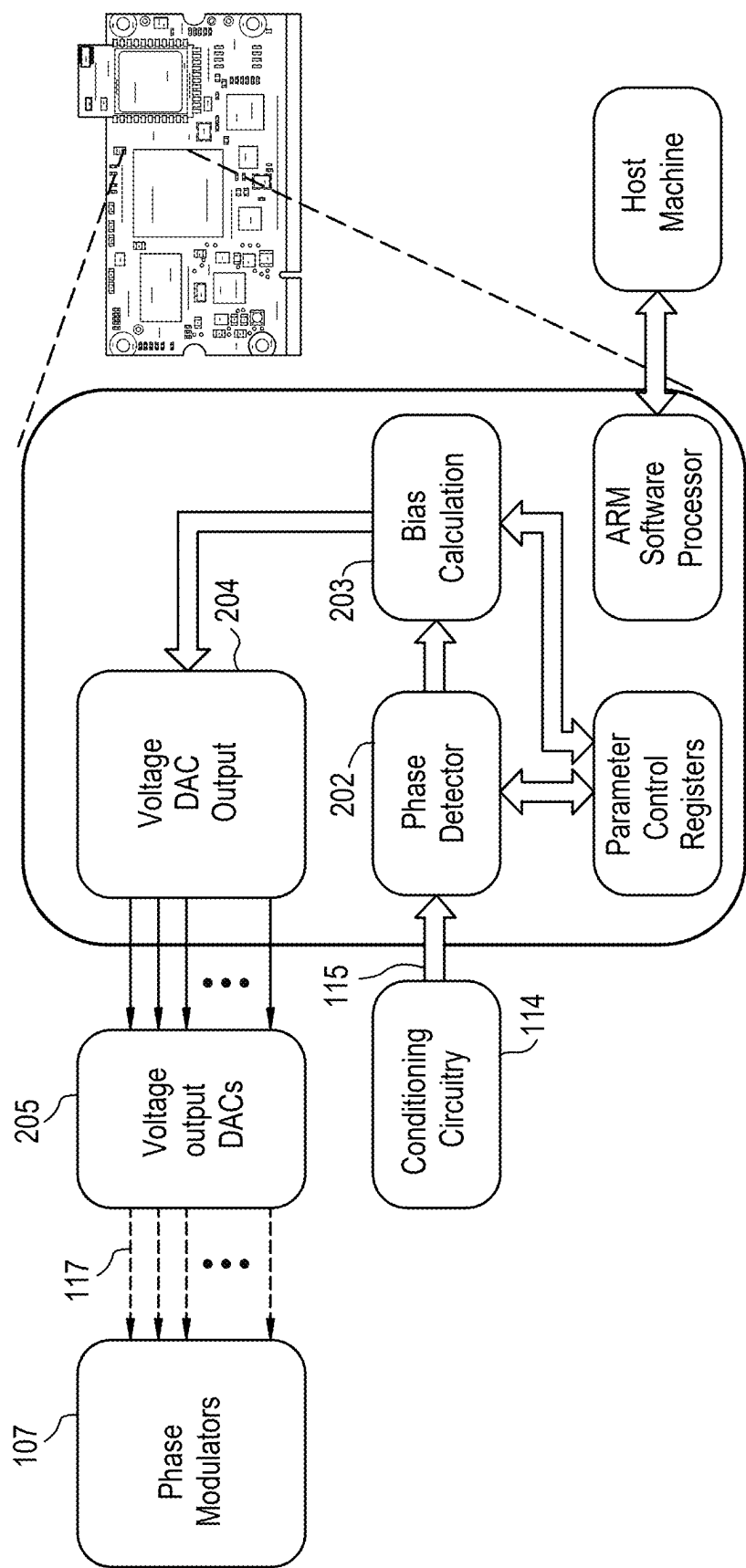


FIG. 2

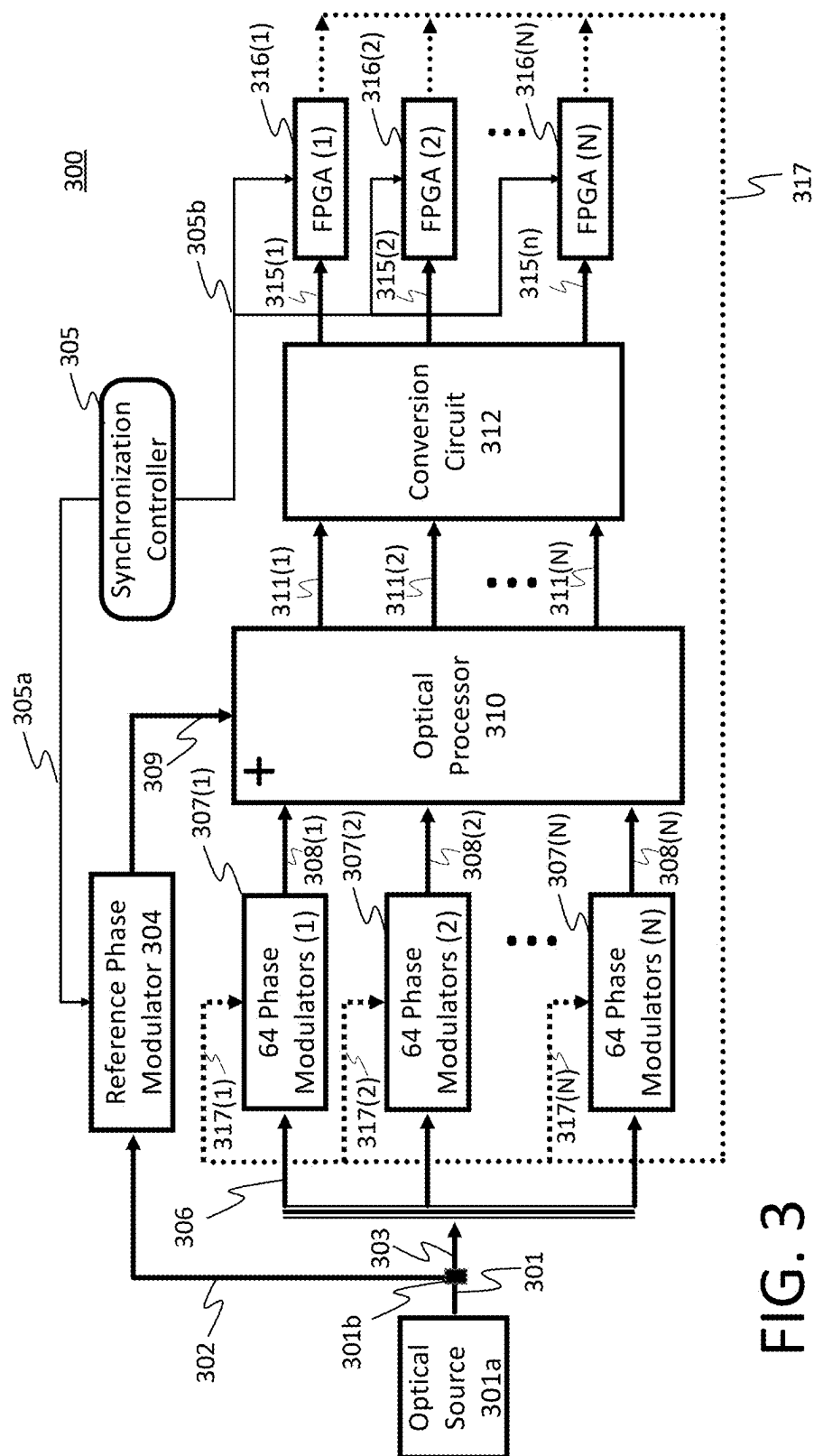


FIG. 3

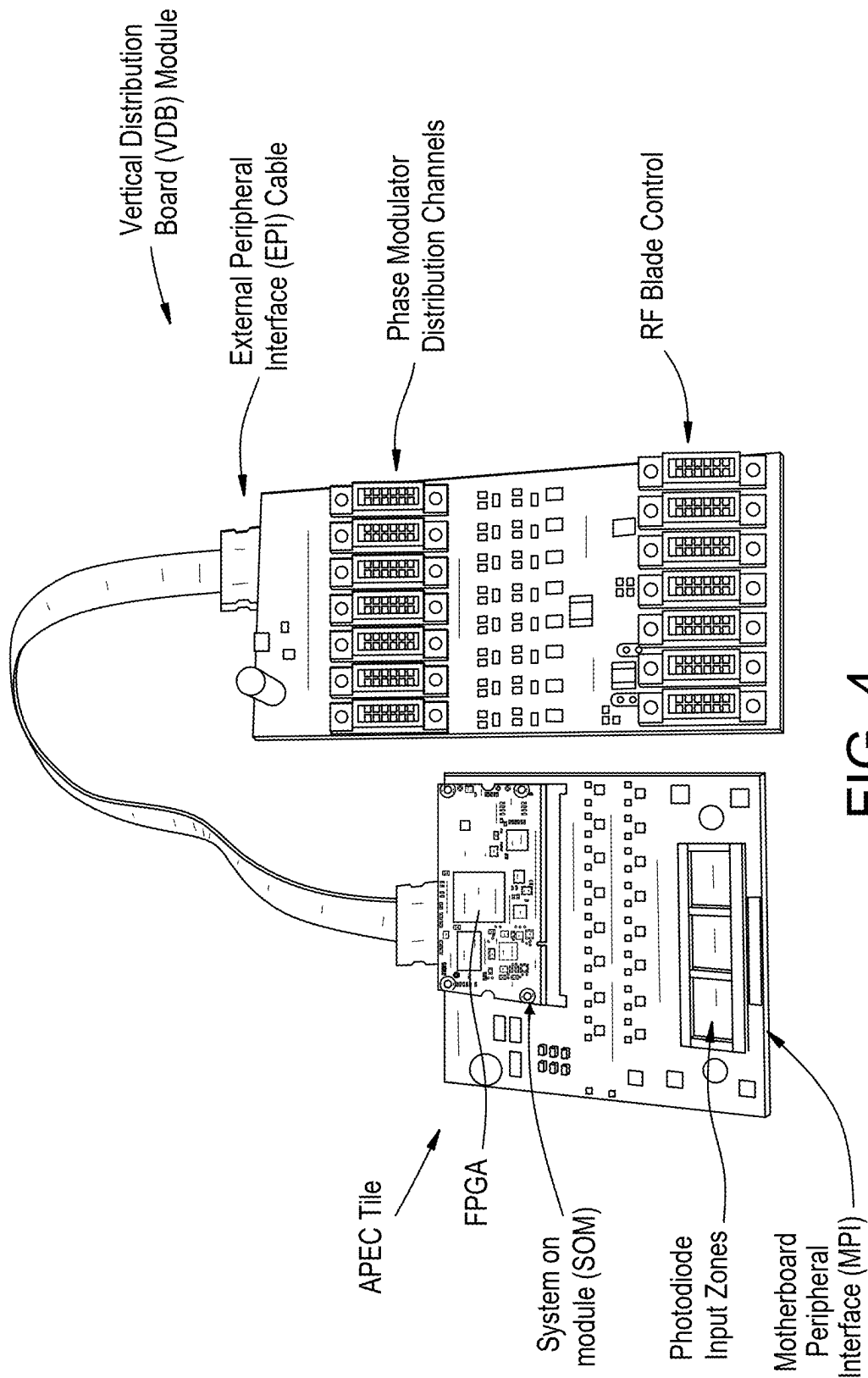


FIG. 4

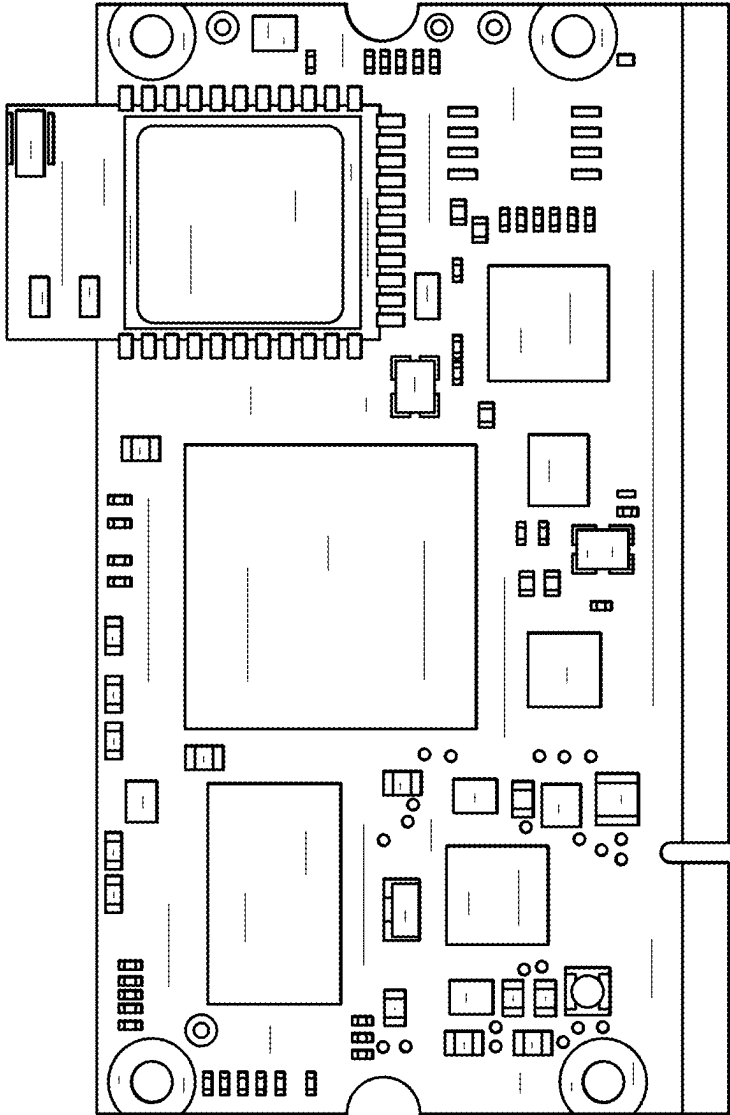


FIG. 5

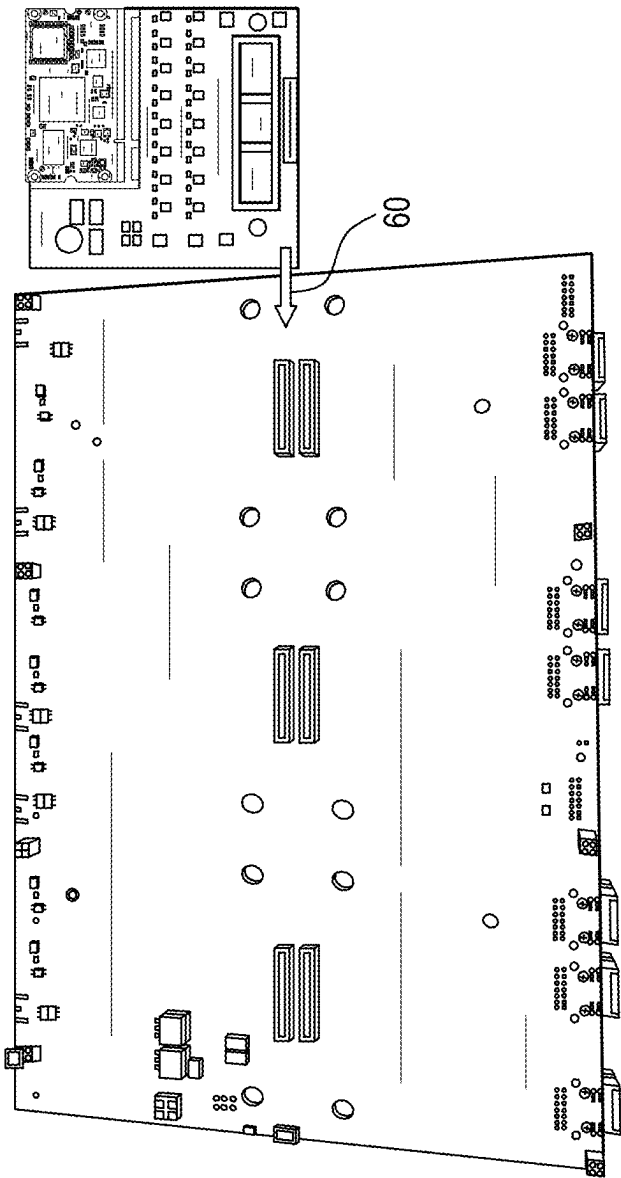
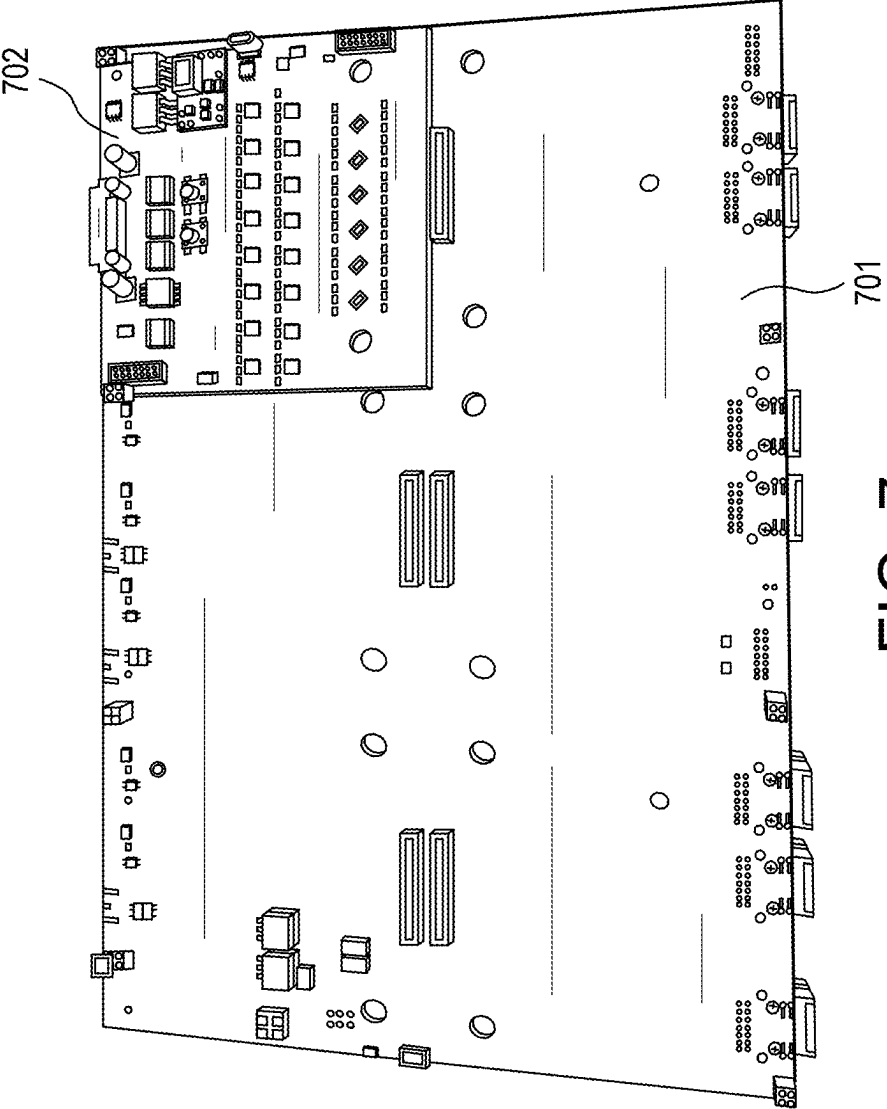


FIG. 6



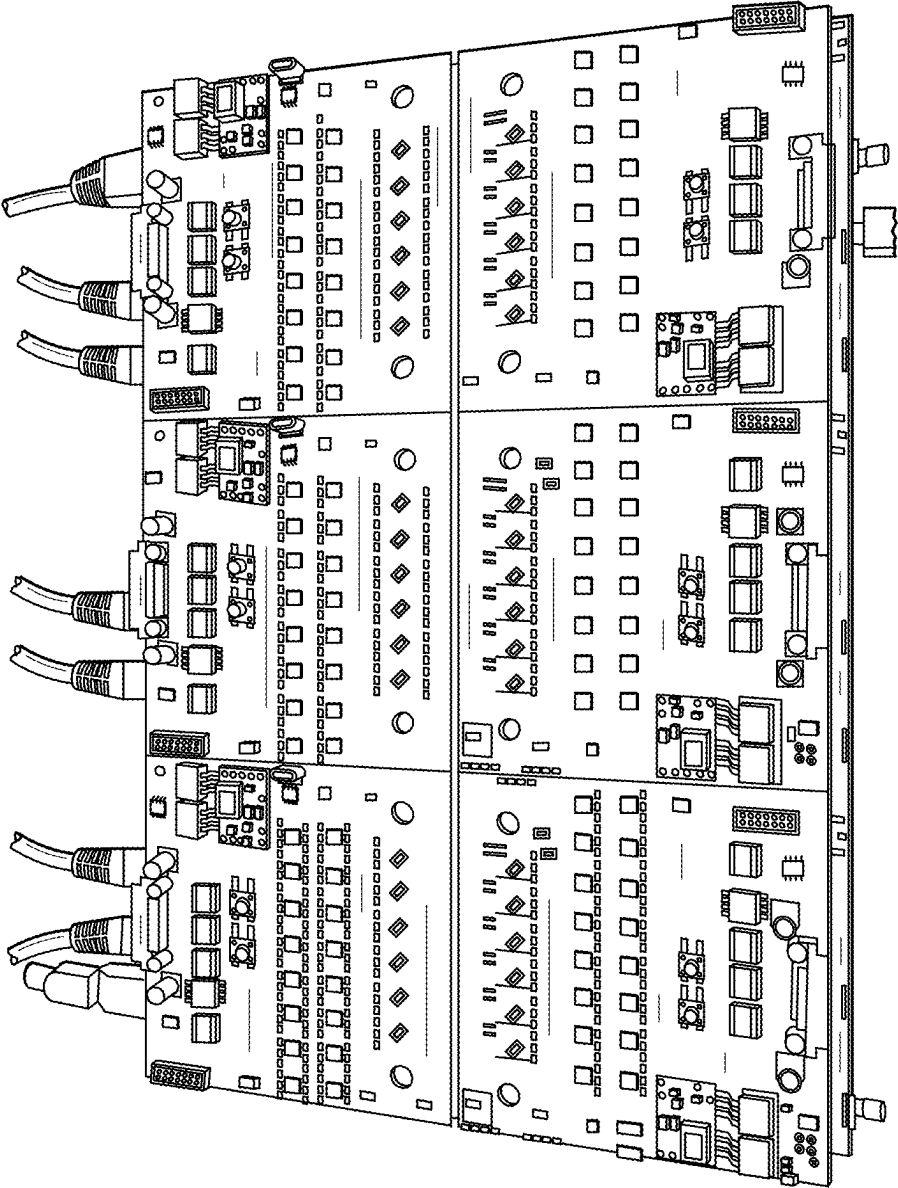


FIG. 8

WIDELY SCALABLE, MODULAR PHASE CONTROL OF OPTICAL CHANNELS

RELATED APPLICATIONS

[0001] This application is a continuation application of U.S. patent application Ser. No. 18/616,753, filed Mar. 26, 2024, which is a continuation application of U.S. patent application Ser. No. 17/890,017, filed Aug. 17, 2022, which is a non-provisional application of Provisional Application No. 63/234,151 filed Aug. 17, 2021, the entire contents of each of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] This disclosure relates generally to digital signal processing, optical coherence, beamforming, and, in particular embodiments to receive signals using phased array antennas.

BACKGROUND

[0003] As novel optical systems transition into the realm of multichannel optics, a fundamental synchronization issue arises of stabilizing the relative phase offset between channels. In such systems, it may be desirable to control each channel's optical phase and essentially "lock" them all to the same pulse or heartbeat for optical synchronicity.

[0004] With the multichannel systems scaling in size, they may require optical phase control that scales commensurately to support a large, or theoretically unlimited, number of channels. One example optical-channel control functionality may be based on the system described in U.S. Patent Pub. No. 2012/0014699, Synchronizing Phases of Multiple Optical Channels, hereby incorporated by reference in its entirety. U.S. Patent Pub. No. 2012/0014699 discloses a method, in which one may extract, process, and control the phases of many optical channels. However, the approach disclosed therein may suffer from hardware-based limitations in channel count due to the limit in the maximum capabilities of digital signal processors known as Field Programmable Gate Arrays (FPGAs). Aspects of the present invention address the scaling limitation of that approach by implementing a distributed, yet synchronized, approach, where multiple digital processors, such as FPGAs, operating in unison replace a single FPGA.

SUMMARY

[0005] According to some embodiments, a plurality of digital processors are used to adjust phases in a plurality of phase modulators. The plurality of digital processors may receive a periodic pulse, or heartbeat signal, from a synchronization controller in order to control the digital processors. The synchronization controller may output an additional signal used to determine and to control the phase of the signals output from the plurality of phase modulators.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] For a more complete understanding of this disclosure, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

[0007] FIG. 1A illustrates a schematic overview of a prior system.

[0008] FIG. 1B illustrates a small sinusoidal signal, its amplified variant, and its discretized calculation of the amplified sinusoid's zero-crossings.

[0009] FIG. 1C illustrates a method in which the phase offset of each optical channel may be digitally captured with respect to a counter reset.

[0010] FIG. 1D illustrates a calculation to determine a phase error and compensation value based on captured offsets denoted in FIG. 1C.

[0011] FIG. 2 illustrates a high-level block diagram of a closed-loop system implemented with an FPGA.

[0012] FIG. 3 illustrates a schematic overview of a system for synchronizing phase of multiple optical signals, according to an example embodiment.

[0013] FIG. 4 shows an implementation of a 64-channel subset system comprising a single APEC Tile, and a connection to a 64-channel phase distribution PCB.

[0014] FIG. 5 shows an iWave G28M FPGA System-on-Module, which may be used to recover phase errors and calculate compensation values.

[0015] FIG. 6 shows a photodiode distribution motherboard and illustrates a connection point to an APEC Tile.

[0016] FIG. 7 shows an APEC Tile mated to a photodiode distribution motherboard using the connection point.

[0017] FIG. 8 shows an implementation of a 384-channel system comprising of 6 APEC Tiles and a motherboard distributing 64 photocurrents to each APEC Tile.

DETAILED DESCRIPTION

[0018] The present disclosure now will be described more fully hereinafter with reference to the accompanying drawings, in which various exemplary implementations are shown. The invention may, however, be embodied in many different forms and should not be construed as limited to the exemplary implementations set forth herein. These example implementations are just that—examples—and many implementations and variations are possible that do not require the details provided herein. It should also be emphasized that the disclosure provides details of alternative examples, but such listing of alternatives is not exhaustive. Furthermore, any consistency of detail between various examples should not be interpreted as requiring such detail—it is impractical to list every possible variation for every feature described herein. The language of the claims should be referenced in determining the requirements of the invention.

[0019] An apparatus comprising a digital processor, such as an FPGA (field-programmable gate array), phase modulators, a laser, an optical processor, electronic conditioning circuitry, and synchronization circuitry may be utilized to coherently align the phases of multiple optical channels.

[0020] Coherent phase alignment of optical channels proves to be a challenge due to the nature of minor environmental acoustic and thermal perturbations in fiber optic cables inducing a phase change in the optical signal carried by the cable. An apparatus may provide a way to compensate for those environmental acoustic and thermal perturbations.

[0021] In U.S. Patent Pub. No. 2012/0014699, published on Jan. 12, 2012, an example method is disclosed wherein an FPGA is used to determine phase errors and compensation values for each respective optical channel.

[0022] As illustrated in FIG. 1A, which represents a conventional system and method, output 101 of an optical source 101a is split into two paths, one path referred to as the reference path 102, and the other referred to as the carrier

path **103**. The reference path is phase modulated by reference phase modulator **104** using a periodic right-triangle (sawtooth) voltage function **105a**, as shown for example in FIG. 2 of U.S. Patent Pub. No. 2012/0014699 or in the top panel of FIG. 1C. The modulating function is generated by the synchronizer **105**, which may be a synchronizer circuit. Synchronizer **105** is controlled by the FPGA **116**. That is, the FPGA will send a control signal (shown by the arrow **105c**) to the synchronizer **105** to cause the synchronizer to transmit the voltage function **105a** to the reference phase modulator **104**. Thus, the system of FIG. 1A is set up to only allow for using one FPGA, which limits the system capabilities. The carrier path is further split as in **106** and propagated into multiple phase modulators **107**. Each phase modulator may heterodyne the carrier signal with an RF signal to encode data into the optical domain. The phase-modulator outputs **108**, and the modulated reference signal **109** are recombined in an optical processor **110** to create combined optical signals **111**, each combined optical signal corresponding to a phase modulator of **107**. The combined optical signals **111** are each an optical waveform that is synthesized through interference between the output **109** of the reference phase modulator **104** and the outputs **108** of the phase modulators **107**. Each combined optical signal **111** may be described as an interference signal. The combined optical signals **111** are then converted into electrical currents **113** by photodiodes of the photodiode array **112** such that each phase modulator has a corresponding electrical current that carries optical phase information in the form of the phase of the sinusoidally varying photocurrent. Examples of these signals can be seen in U.S. Patent Pub. No. 2012/0014699, FIGS. 3A-3E, and FIG. 4, top sinusoidal signal. For example, in U.S. Patent Pub. No. 2012/0014699, FIG. 3A shows an example sawtooth voltage function, FIG. 3B shows an example modulated reference signal **109**, FIG. 3C shows an example phase-modulator output **108**, FIG. 3D shows a combined signal (combining modulated reference signal **109** with a phase-modulator output **108**), and FIG. 3E shows just the interference pattern resulting from the combined signal (e.g., a sinusoidal signal). FIGS. 3A-3E of U.S. Patent Pub. No. 2012/0014699 are optical signals.

[0023] The electrical signals, or electrical currents **113** are conditioned, or converted in conditioning circuitry **114** using transimpedance amplifiers and comparators (e.g., as depicted in FIG. 5 of U.S. Patent Pub. No. 2012/0014699) to yield conditioned waveforms **115** such that the photocurrent is converted to a voltage, and inflection points are converted to rising and falling edges of a square wave pattern, which can be considered a digital signal, converted from analog. For example, in FIG. 4 of U.S. Patent Pub. No. 2012/0014699, the lower square wave depicts an electrical signal, after being converted by photodiode array **112**, for example, from the optical signals input to photodiode array **112**, and being further processed by conditioning circuitry **114**. Embodiments of this conditioning circuitry may entail those listed in U.S. Patent Pub. No. 2012/0014699.

[0024] An example conditioning process is illustrated in FIG. 1b, where the left panel represents the photocurrent at the output of a photodiode **113**, the middle panel represents the voltage at the output of a transimpedance amplifier, and the right panel represents the output of a comparator. Equivalently, the comparator may be thought of as converting the continuous sine wave to a discretized, two-level

waveform, and other embodiments of this invention may use a 1-bit analog-to-digital converter (ADC).

[0025] By virtue of having discretized outputs, a digital processor such as an FPGA **116** can now sample the data and interpret the inflection points as a change of state from a “zero” to a “one” on the rising-edge inflection points, or vice-versa for the falling inflection points. The timing of the rising/falling edges is then correlated in the FPGA with a global counter associated with a data-sampling clock, and the counter resets for every cycle at the falling edge **132** of the sawtooth function driving the reference phase modulator **104**, as illustrated in FIG. 1C, where the Ref. Saw-tooth represents the synchronizer output **105a** driving the reference phase modulator **104**, the Counter Reset **133** represents the start of a new periodic cycle of the sawtooth, the Comparator Output **134** represents a conditioned waveform **115** from the combination of the reference and an individual carrier channel, the Global Counter **135** illustrates digital clock cycles to measure the timing of the inflection points of the Comparator Output **134**, the Offset1 **136** and Offset2 **137** variables correspond to latched values of Global Counter, and Edge Type **138** corresponds to whether the Comparator Output **134** had an initial rising or falling edge.

[0026] By virtue of resetting the phase counter with every cycle of the sawtooth, the phase error and compensation values of each carrier channel of the array can be extracted in a way that is analogous to that explained in U.S. Patent Pub. No. 2012/0014699 and corrected for with an error voltage adjustment in the carriers’ phase modulators **107**. This error voltage is obtained by pipelining a calculation between the global phase counter and the inflection points of each carrier channel input **115** into the FPGA, whose calculation data path is illustrated in FIG. 1D. An FPGA implements this calculation data path. For example, as shown in FIGS. 1C and 1D, the average between Offset2 **137** and Offset1 **136** is taken, and a 180 degree phase shift is either added or is not added, depending on the Edge Type **138**. The Edge Type **138** is determined by whether Comparator Output **134** had an initial rising edge or an initial falling edge. The resultant value of the addition, or lack thereof, is compared to a setpoint and proportionally multiplied to calculate an appropriate error voltage **117** per channel. This error voltage may be shifted by an equivalent error voltage resulting in a 360 degree phase shift either in the positive or negative direction in order to remain within the same range, or ranges to maintain the error voltages **117** within 360 degrees of range, or any integer multiple of 360 degrees, depending on the supply voltage of the error voltage applied to the phase modulator. FIG. 2 illustrates the digital calculation data path of the discretized channel input. Conditioning circuitry **114** feeds a phase detector that calculates the Offset1 **136** and Offset2 **137** values of FIG. 1C, a bias calculation block **203**, whose function is illustrated in FIG. 1D, and a parallel DAC output **204** that drives a digital state machine for high-speed control of multiple voltage-output DACs **205** to supply a bias based on the calculated compensation values to each individual phase modulator for fine-grain phase control. More generally, if the start (e.g., falling edge) of the sawtooth is synchronized with the counter reset pulse, and the global counter is counting an amount of time between the counter reset and the time at which the comparator output signal changes from 0 to 1 and from 1 to 0, then a point of maximum and minimum interference in the signals **111** can be determined, and so a

phase shift needed to make the reference signal in reference path **102** and the signal **106** going into each phase modulator **107** be in phase with other, and to make the reference signal in reference path **102** and the signal **108** coming out of each phase modulator **107** be in phase with each other or have a pre-set phase relationship with or phase difference from each other that remains constant over time, can be determined. Accordingly, the voltage-output DACs provide error voltages **117**, shown in FIG. 1A, that bias phase modulators **107** to adjust the phase of signals **108** output from phase modulators **107**. Further details of this calculation and data path according to some examples are disclosed in U.S. Patent Pub. No. 2012/0014699 and may be used to implement aspects of the present invention.

[0027] With the stated description of the prior art, an embodiment of this invention is now presented such that the phases of each optical phase modulator can be controlled in real-time such that their phase errors can be locked or discretely phase shifted and then locked to realize a known phase offset of each optical channel with respect to a sawtooth voltage function. This optical phase lock is implemented with a closed-loop optoelectronic network including a plurality of digital processors, such as FPGAs, all controlled by a separate digital processor, described as a synchronization controller. The synchronization controller acts independently and is not under the control of an FPGA such as FPGA **116** of FIG. 1A. Contrary to the conventional art where a digital processor controls its own timing as well as the timing of a synchronizer, the synchronization controller according to embodiments of the present invention sends control signals to a plurality of digital processors, to control those digital processors simultaneously. Further details are described in connection with FIG. 3 and later figures, as discussed below.

[0028] FIG. 3 illustrates a schematic overview of a system for synchronizing phase of multiple optical signals, according to an example embodiment. As illustrated in FIG. 3, a system **300** includes an optical source **301a**, a splitter **301b**, a plurality of phase modulators **307** organized in phase modulator groups **307(1)-307(N)**, a reference phase modulator **304**, a synchronization controller **305**, an optical processor **310**, a conversion circuit **312**, and a plurality of digital processors **316(1)-316(N)**. As is traditional in the field of the disclosed technology, features and embodiments are described, and illustrated in the drawings, in terms of functional blocks, units and/or modules. Those skilled in the art will appreciate that these blocks, units and/or modules are physically implemented by electronic (or optical) circuits such as logic circuits, discrete components, microprocessors, hard-wired circuits, memory elements, wiring connections, and the like, which may be formed using semiconductor-based fabrication techniques or other manufacturing technologies. In the case of the blocks, units and/or modules being implemented by microprocessors or similar, they may be programmed using software (e.g., microcode) to perform various functions discussed herein and may optionally be driven by firmware and/or software. Alternatively, each block, unit and/or module may be implemented by dedicated hardware, or as a combination of dedicated hardware to perform some functions and a processor (e.g., one or more programmed microprocessors and associated circuitry) to perform other functions. Also, each block, unit and/or module of the embodiments may be physically separated into two or more interacting and discrete blocks, units

and/or modules without departing from the scope of the inventive concepts. Further, the blocks, units and/or modules of the embodiments may be physically combined into more complex blocks, units and/or modules without departing from the scope of the inventive concepts.

[0029] The optical source **301a** may be, for example, a laser or another light source, which may output an optical source signal **301**, which may be a laser light, or other light having a particular frequency. The splitter **301b** may be any known optical splitter (e.g., a partially reflective glass, a glass prism or cube, or a fiber splitter, or a fiber directional coupler, etc.), connected and configured to split the optical source signal **301** into multiple signals. For example, one signal output from the splitter **301b** is a reference signal **302**, and another signal output from the splitter **301b** is a carrier signal **303**. Reference signal **302** may be input to a reference phase modulator **304**. Carrier signal **303** may be further split into a plurality of signals **306**, which may be input into phase modulators **307**. The combination of splitter **301b** and additional splitters for forming a plurality of carrier signals input into the phase modulators **307** may be described together as a splitter, which splits the optical source signal **301** into a plurality of signals **306** including a plurality of phase modulator input signals including the carrier signal, each of which may be described as a carrier signal, and a reference signal. The carrier signals and the reference signal may be synchronized to initially have the same frequency and phase.

[0030] Reference phase modulator **304** is connected to receive reference signal **302** from optical source **301a** as well as a signal **305a** from synchronization controller **305**, and to output a modulated reference signal **309**. For example, the signal **305a** may be a voltage function, which may have a periodic pattern, such as a sawtooth pattern such as discussed previously in connection with FIG. 1C. The voltage function may represent phase versus time. In operation, reference phase modulator **304** receives a voltage function **305a** and reference signal **302**, and modulates the reference signal using the voltage function, as discussed for example, in connection with FIG. 1C. The modulated reference signal **309** is transmitted to an optical processor **310**. For example, the voltage function **305a** may be a signal that causes the phase of the modulated reference signal, to linearly vary by a particular amount (e.g., 1 degree) over time (e.g., at each clock cycle or step of the voltage function). Therefore, at a first time, the phase of the modulated reference signal **309**, which initially has the same frequency as the reference signal **302** and the carrier signal **303**, may be shifted by one degree in phase in a particular direction, at a second time, the phase of the modulated reference signal **309** may be shifted by two degrees in phase in the same direction, etc.

[0031] Synchronization controller **305** is connected to transmit the signal **305a** (e.g., a voltage function) to reference phase modulator **304** and is connected to transmit a signal **305b** to a plurality of digital processors **316(1)-316(N)**. The signal **305b** may be, for example, a voltage function start indicator signal, that includes, for example, a pulse that indicates the start of the voltage function being transmitted as signal **305a** to the reference phase modulator. The start may be, for example, a value that causes the phase of the modulated reference signal **309** to be shifted 0 degrees with respect to the reference signal **302**. The voltage function start indicator signal may be described as a heartbeat signal,

or a periodic pulse signal. Different from prior art systems such as discussed previously, the heartbeat signal **305b** is transmitted from synchronization controller **305** to the digital processors **316(1)-316(N)**. The heartbeat signal **305b** may therefore control the digital processors **316(1)-316(N)** in a manner to be described in more detail below. The heartbeat signal **305b** may be a similar signal to that described in connection with FIG. 1C (signal **133**) above, and may be a counter reset signal that resets a counter that begins counting at the beginning of a cycle of the voltage function **305a**. Synchronization controller **305** may be, for example, a digital processor such as an FPGA. The heartbeat signal **305b** may be transmitted simultaneously and synchronously to each digital processor **316**.

[0032] Phase modulators **307** may be organized into groups, such as a first group **307(1)**, a second group **307(2)**, and additional groups up to an Nth group **307(N)**. In some embodiments, N may be an integer having a value from 2 to 16. However, N is not limited as such, and may be greater than 16 (e.g., 32, 64, or higher). In one embodiment, each phase modulator group **307(1)-307(N)** includes 64 phase modulators, and each phase modulator is connected to receive a carrier signal as an input and outputs a signal **308** described as a phase modulator output signal or phase modulator output, or alternatively described as an optical input signal. Phase modulator output signals **308** may be grouped into groups **308(1)-308(N)**, which correspond to the phase modulator groups **307(1)-307(N)**.

[0033] Each phase modulator **307** may receive as input a carrier signal and an information signal (not shown), and converts the information signal to the optical domain using the carrier signal. For example, the information signal may be an RF signal, or other signal. As an example, each phase modulator **307** may heterodyne the carrier signal **306** with an RF signal or other information signal to encode data into the optical domain. Each phase modulator **307** may be configured to perform phase adjustment of the carrier signal and modulation of the information signal using the phase-adjusted carrier signal. Each information signal may include, for example, a plurality of bits of information, encoded in the RF or other domain, or may be an analog signal having information encoded therein. Each phase modulator **307** may also receive a respective adjustment signal **317**. The adjustment signals **317** may similarly be organized into a plurality of groups **317(1)-317(N)**. Each adjustment signal **317** may be used along with a respective information signal and carrier signal to output a phase modulator output signal **308**. The adjustment signals are output from the digital processors **316(1)-316(N)**, and are described in greater detail below.

[0034] Optical processor **310** is connected to receive the modulated reference signal **309** and the phase modulator output signals **308**, and is configured to, for each phase modulator output signal **308**, combine the phase modulator output signal **308** with the modulated reference signal **309**. For each phase modulator output signal **308** combined with the modulated reference signal **309**, a respective interference signal **311** is generated. The interference signal **311** reflects whether the modulated reference signal **309** is in phase with the respective phase modulator output signal **308**, and may be a periodic signal (e.g., sine wave) having the same frequency as the frequency of a sawtooth voltage function. For example, the magnitude of the interference signal **311** may change for each degree of variation in the voltage

function **305a**, to have the same frequency as the voltage function **305a**. An example of this can be seen in FIG. 3E of U.S. Patent Pub. No. 2012/0014699. Each phase modulator output signal **308** input to optical processor **310** (e.g., optical input signal) may therefore have a corresponding respective interference signal **311** output from the optical processor **310** (e.g., optical output signal). The interference signals may vary depending on differences in the phase of the different phase modulator output signals **308**. The phase differences may be pre-set by a predetermined offset for some phase modulators **307**, but may additionally include a component due to environmental factors that is not intended. The phase of the phase modulator output signals **308** can be further adjusted by phase modulators **307**, as will be described below.

[0035] Conversion circuit **312** is connected to receive the interference signals **311**, which in one embodiment are analog optical signals, and convert each interference signal **311** to the digital electrical domain. For example, conversion circuit **312** may include a plurality of photodiodes formed in a photodiode array, and may include one or more comparators. For example, conversion circuit **312** may include components such as depicted in FIG. 5 of U.S. Patent Pub. No. 2012/0014699. As a result of the optical-to-electrical conversion, and analog-to-digital conversion, in one embodiment, for each interference signal, a square wave is generated, which has rising and falling edges that correspond to zero crossings of the interference signal. This signal is then output from the conversion circuit **312** and is input to a digital processor **316**. This is only one example, however. The rising and falling edges need not correspond to the zero crossing if a different type of circuit is used to convert from analog to digital. For example, another circuit could be used that causes the rising and falling edges of the square wave to correspond to a peak and valley of the interference signal.

[0036] As shown in FIG. 3, the conversion circuit **312** can receive interference signals from the optical processor **310** as N groups of interference signals **311(1)-311(N)**, and can respectively output converted signals as N groups of converted signals (e.g., resulting digital signals) **315(1)-315(N)**. Each group of converted signals can then be input into a single respective digital processor **316** from among digital processors **316(1)-316(N)**.

[0037] Digital processors **316** may be, for example FPGAs, where each FPGA is connected to receive a group of converted signals **315** from groups **315(1)-315(N)** and to output a group of adjustment signals **317** from adjustment signal groups **317(1)-317(N)**. However, digital processors **316** can be other types of processors, such as complex programmable logic devices (CPLDs), digital signal processors (DSPs), or application-specific integrated circuits (ASICs).

[0038] A method is now disclosed such that multiple digital processors, such as FPGAs, may be operated in parallel to overcome the limitations of a single electronic system. This method is illustrated in FIG. 3, where compared to FIG. 1, a photodiode array **112** and conditioning circuitry **114** is replaced with a conversion circuit **312**, which may be a photodiode array and breakout module, which distributes a group of photocurrents (e.g., digital signals that correspond to outputs from optical processor **310**) to each individual digital processor (e.g., FPGA module). Each digital processor **316** is now responsible for its

own subset, or group, of phase modulators **307**. In one embodiment, each group of phase modulators **307** may include 64 phase modulators. The synchronization controller **305** is now utilized to generate a reference heartbeat **304** to be distributed and delay-matched to each digital processor **316** for synchronicity. Each digital processor **316** now implements its own closed control loop algorithm similar to the one illustrated in FIG. 2.

[0039] As discussed previously, a plurality of adjustment signals **317** may be sent from each digital processor **316** to each group of phase modulators **307(1)-307(N)**, and each phase modulator **307** can use the received adjustment signal **317** to shift the phase of the output signal to result in a phase modulator output signal **308**. Thus, each phase modulator **307** includes a circuit configured to change the phase of the output signal **308** by an amount corresponding to the adjustment signal **317**. Each phase modulator **307** may initially be set, for example by a digital processor **316**, so that the phase modulator output signal **308** has a pre-set phase relationship to the initial carrier signal **303**. For example, the pre-set phase relationship could be that the two signals are in phase, or can be that the two signals have a predetermined phase difference between them, described as a predetermined offset. In some applications, the plurality of digital processors **316** can be configured to cause the plurality of phase modulator output signals **308** to have pre-set phase relationships with respect to each other that may not be in phase (e.g., a blazed phase profile, or a parabolic phase profile). The adjustment signal **317** can be set by the digital processors to result in this phase profile for the phase modulators **307**. Each adjustment signal may be, for example, a voltage that biases the phase modulator **307** to adjust the phase of the signal being output. The adjustment signals **317** may be generated by the digital processors **316**, for example, using hardware and/or software that calculates a voltage according to an equation such as shown and discussed in connection with FIG. 1D, where offsets **1** and **2**, a global counter and edge type information, and a setpoint (Set[n]) are used as inputs, and a control signal (e.g., adjustment signal **317**, which may be a voltage value) is output. The adjustment signal **317** may therefore be based on both a set point, which for each phase modulator **307** may correspond to a pre-set phase relationship between the carrier signal **303** and the phase modulator output signal **308**, and offsets that may result from environmental factors.

[0040] In one embodiment, the plurality of digital processors **316** are configured to repeat the process of outputting the adjustment signals **317** over time, in order to control the phase of the respective phase modulator outputs **308(1)-308(N)** so that the plurality of phase modulator outputs are in phase with each other or so that relative phase offsets between different phase modulator outputs are maintained to be the same, and thus remain stable, over time. This can lock the phase relationship between the phase modulator outputs over time, and allows for predictable processing of the plurality of phase modulator outputs, even while environmental factors may be changing over time. According to some embodiments described herein, a digital processor may calculate a compensational voltage to shift the phase of each phase modulator of a plurality of phase modulators independently. A plurality of digital processors may be used to shift the phase of a plurality of groups of phase modulators, and the plurality of digital processors may be controlled by a single controller. This may happen in real-time

to create a “phase-lock”, as well as tune individual phase modulators to a specific phase to remain locked in a predetermined relationship with other phase modulators.

[0041] A unit module according to some embodiments is shown in FIG. 4. The individual unit module, allows for sampling and synchronizing the phases of 64 individual optical channels using a single FPGA, pictured in FIG. 5.

[0042] The phase-control functionality of a 64-channel unit module of this apparatus can be expanded into multiple unit modules **316**, as illustrated in FIG. 3, where modules are denoted as FPGA (**1**) to FPGA (**N**). By increasing the number of optical carriers and the width of the reference source **109**, one may scale to as many channels as can be optically processed. A photodiode array should be capable of converting as many channels as an optical processor can provide into electrical currents, at which point the system size is limited only by the circuit board footprint limit, and the limit of optical-component sizes.

[0043] Given that one may make a PCB as large as 3x3 ft in a standard fabrication order or by combining multiple PCBs, a PCB may create a very large footprint for a photodiode array, potentially capable of supporting many thousands of channels. Given that free-space optics can be made as large as one may like, limitations on the optical processing channel count are removed. By having a single lightweight master microcontroller, which may be a digital processor such as an FPGA module, generate and control the sawtooth and a global phase “counter reset” pulse to all other digital processors, which may also be FPGA modules, which FPGA modules are sampling and correcting channels, one may use as many 64-channel modules as needed to sample, correct, and control all the optical phases. Synchronous application of the phase counter reset results in a ubiquitous, modular approach, where each FPGA module only knows its own realm of 64 channels and the system can be scaled to an unlimited number of channels.

[0044] A scaled implementation having 384 channels according to one embodiment is now described. According to one embodiment of this invention, to control the optical phases of 384 different channels, the system was implemented with 6 Analog-to-Digital Phase Error Correction (APEC) tiles. FIG. 6 shows a “motherboard” (e.g., a printed circuit board) with a single APEC tile to its right. Arrow **60** shows how the APEC tile connects to the “motherboard.” Each APEC tile consists of an iWave G28M System-on-Module as a digital processor, an entry point for photocurrents or a photodiode array, and photocurrent conditioning circuitry (e.g., conversion circuitry), and a means of applying phase error correction signals to Digital-to-Analog converters for phase modulator interfacing. FIG. 7 demonstrates an APEC tile **702** fitted into said motherboard **701**. The motherboard provides a structure that “fans-out” the input channels into groups of 64, for each APEC tile to individually process. FIG. 8 illustrates a fully integrated 6-APEC Tile array to ultimately process 384 optical channels. Each APEC tile feeds its processed values to a phase distribution board over a data cable, as shown in FIG. 4. This board will then apply those error voltages into phase modulators to individually lock and control the phases of every optical channel fed to the motherboard. The motherboard may include the synchronization controller mounted thereon, separately from the APEC Tiles, such as a digital processor.

In one embodiment, the synchronization controller is mounted on the backside of the motherboard (not seen in FIGS. 6-8).

[0045] This system theoretically has an ability to control an unlimited number of optical channels using a low-cost, low-SWAP (Size, Weight, and Power) implementation. The method described here has been proven to work for as many channels as one can supply to the APEC tiles. For example, $64 \times N$ input channels may be processed, where N is the number of APEC tiles.

[0046] This invention may be used to synchronize phases both within a system and external to the system. The synchronized “counter reset” pulse may be distributed to as many APEC tiles, or systems, as one may need. Multiple systems can be synchronized to one effective heartbeat, resulting in widespread synchronicity.

[0047] Terms such as “the same” or phrases such as “maintained to be the same” are intended to include minor variations that do not otherwise affect the operation of the system. The term “substantial” or “substantially” may be used to reflect this meaning.

[0048] The following publications also are incorporated by reference in their entirety and provide details of systems and methods in which this invention may also be implemented:

[0049] Dillon, Thomas E., et al. “Passive millimeter wave imaging using a distributed aperture and optical upconversion.” *Millimetre Wave and Terahertz Sensors and Technology III*. Vol. 7837. International Society for Optics and Photonics, 2010.

[0050] C. A. Schuetz, J. Murakowski, G. J. Schneider and D. W. Prather, “Radiometric Millimeter-wave detection via optical upconversion and carrier suppression,” in *IEEE Transactions on Microwave Theory and Techniques*, vol. 53, no. 5, pp. 1732-1738, May 2005, doi: 10.1109/TMTT.2005.847106.

What is claimed is:

1. An apparatus for controlling phase of a plurality of optical signals, the apparatus comprising:

- a plurality of phase modulators, each configured to be connected to receive an optical carrier signal, and based on the optical carrier signal to transmit an output signal, thereby resulting in a plurality of respective phase modulator output signals as optical input signals;
- a synchronization controller configured to generate and output a periodic voltage function and a periodic pulse signal; and
- a plurality of digital processors, connected to simultaneously receive the periodic pulse signal output by the synchronization controller, each digital processor configured to output a respective adjustment signal to each respective phase modulator of a respective group of phase modulators, wherein the respective adjustment signals are based on a respective group of interference signals, each interference signal of the group of interference signals generated based on the periodic voltage function and a respective phase modulator output signal, and each respective adjustment signal causing a respective phase modulator to control the phase of its respective phase modulator output signal.

2. The apparatus of claim 1, further comprising:

- a reference phase modulator configured to be connected to receive the periodic voltage function from the synchronization controller and to receive a reference signal

having the same phase and frequency as the optical carrier signal, and configured to modulate the reference signal according to the periodic voltage function to generate a modulated reference signal, and to output the modulated reference signal.

3. The apparatus of claim 2, further comprising:

an optical processor connected to receive the modulated reference signal and the optical input signals and configured to, for each particular optical input signal of the optical input signals, generate an interference signal of the group of interference signals based on the particular optical input signal and the modulated reference signal, thereby generating a respective interference signal for each respective optical input signal,

wherein each interference signal of each group of interference signals is generated by modulating the reference signal with the periodic voltage function to generate a modulated reference signal, and by combining the modulated reference signal with a respective phase modulator output signal.

4. The apparatus of claim 2, further comprising a splitter configured to output the reference signal to the reference phase modulator and to output the carrier signal to the plurality of phase modulators.

5. The apparatus of claim 1, wherein the periodic voltage function is a signal having a sawtooth shape.

6. The apparatus of claim 1, wherein the periodic pulse signal is a voltage function start indicator signal that resets a counter value.

7. The apparatus of claim 1, wherein the synchronization controller is a field programmable gate array (FPGA).

8. The apparatus of claim 1, wherein each digital processor of the plurality of digital processors is an FPGA.

9. The apparatus of claim 8, wherein the FPGAs that form the plurality of digital processors together cause adjustments in the phase modulator output signals so that at least some of the phase modulator output signals are adjusted in phase with respect to the reference signal by a predetermined offset.

10. The apparatus of claim 9, wherein the FPGAs that form the plurality of digital processors are configured to control the phase of the respective phase modulator output signals so that the plurality of phase modulator output signals are in phase with each other or so that relative phase offsets for the different phase modulator output signals are maintained to be the same over time.

11. The apparatus of claim 1, wherein each digital processor is configured to send a plurality of respective adjustment signals to a subset of phase modulators of the plurality of phase modulators.

12. The apparatus of claim 11, wherein each subset of phase modulators includes 2^n phase modulators, where n is between 2 and 7.

13. The apparatus of claim 1, further comprising:

an analog-to-digital converter, configured to receive the interference signals of the group of interference signals, convert each received interference signal from analog to digital and from optical to electrical, and output the resulting digital signals to the plurality of digital processors.

14. The apparatus of claim **1**, further comprising:
a circuit board that includes the synchronization controller mounted thereon, and on which the plurality of digital processors are mounted, separately from the synchronization controller.

15. The apparatus of claim **14**, wherein each digital processor includes a respective circuit board.

16. A method for controlling phase of a plurality of optical signals, the method comprising:

receiving a reference signal and a carrier signal, the reference signal and carrier signal being optical signals having the same phase and frequency;

using the carrier signal to generate a plurality of optical input signals, the plurality of optical input signals including a first group of optical input signals and a second group of optical input signals;

generating a respective interference signal for each particular optical input signal of the plurality of optical input signals, by using each respective particular optical input signal and the reference signal, thereby generating a plurality of interference signals including a first group of interference signals and a second group of interference signals;

converting the plurality of interference signals to digital signals to generate a plurality of digital signals including a first group of digital signals and a second group of digital signals;

inputting the first group of digital signals to a first digital processor, and simultaneously inputting the second group of digital signals to a second digital processor;

transmitting, by the first digital processor, a first group of adjustment signals respectively corresponding to the first group of optical input signals, and transmitting, by the second digital processor, a second group of adjustment signals respectively corresponding to the second group of optical input signals; and

adjusting the phase of at least one optical input signal of the first group of optical input signals based on the first group of adjustment signals, and adjusting the phase of

at least one optical input signal of the second group of optical input signals based on the second group of adjustment signals.

17. The method of claim **16**, wherein adjusting the phase of the at least one optical input signal of the first group of optical input signals, and adjusting the phase of the at least one optical input signal of the second group of optical input signals is repeated over time so that phases of the plurality of optical input signals or relative phase offsets for the plurality of optical input signals are maintained to be the same over time.

18. The method of claim **16**, wherein generating the respective interference signal for each particular optical input signal of the plurality of optical input signals includes modulating the reference signal using a periodic voltage function to generate a modulated reference signal, and using the modulated reference signal to generate the plurality of interference signals.

19. A synchronization controller for controlling phase of a plurality of optical signals in a multichannel optical signal processing apparatus including a plurality of phase modulators and a plurality of digital signal processors, the synchronization controller comprising:

an independent digital processor configured to:

generate and output a periodic pulse signal to the plurality of digital signal processors in order to control a plurality of respective phase adjustment signals generated by the plurality of digital signal processors; and

generate and output a periodic voltage function to a reference phase modulator, the periodic voltage function controlling modulation of a reference signal used to generate respective interference signals when combined with respective optical signals controlled by the plurality of respective phase adjustment signals.

20. The synchronization controller of claim **19**, wherein the periodic pulse signal is a voltage function start indicator signal.

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